

Linear Systems on Graphs

Controllability of Structured Networks

Chapter 1

Introduction

Systems and Models · From One Plant to Networks · Book Outline

Department of Control & Instrumentation Engineering
Pukyong National University, Busan, Korea

pnj@pknu.ac.kr

Chapter 1 Contents

1.1 Physical Systems and Models

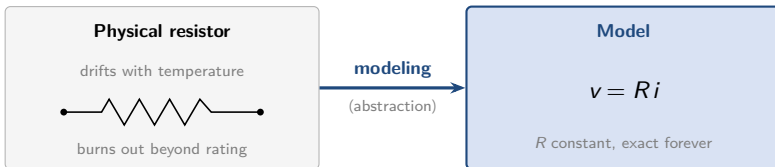
1.2 From One Plant to Networks

1.3 Book Outline

From Physical System to Model

CONCEPT Modeling: Physical System → Model

- ▶ Engineering **analysis** works on a **mathematical model**, not the real system
- ▶ **Resistor**: the real part drifts with temperature and burns out beyond its rating
→ the model keeps a **constant** R
- ▶ **Robot link**: the real link flexes under load → the model keeps it **perfectly rigid**
- ▶ Every later result is bounded by **model quality**: **the model is not reality**

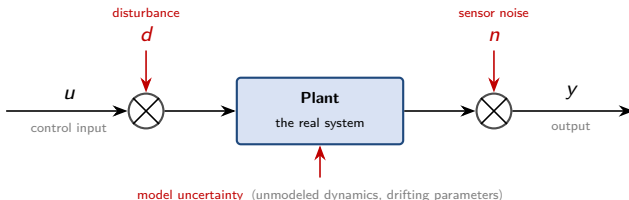


▶ Convention of This Course

- ▶ From now on, "system" always means a **model of a physical system**
- ▶ Modeling is assumed done, and the starting point is the model itself

The Modeling Gap: Disturbance and Uncertainty

- ▶ The plant is driven by the control input u we choose and a **disturbance** d we neither choose nor measure
- ▶ The output is read through a sensor, so the measurement carries **noise** n as well
- ▶ A nonlinear plant is approximated by **linearization** about an operating point, giving the model $\dot{x} = Ax + Bu$ used from here on

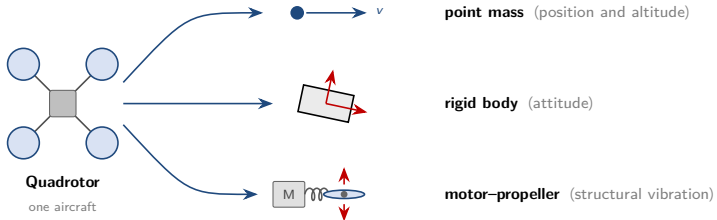


[The modeling gap] No model matches the real system exactly. A model only has to keep the error small over the **operating range of interest**.

One Physical System, Three Models

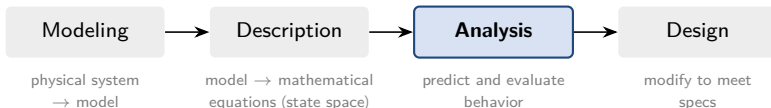
EXAMPLE Quadrotor

- ▶ To a **path planner**: a **point mass**
- ▶ To an **attitude controller**: a **rigid body**
- ▶ To a **vibration analyst**: **coupled motor–propeller dynamics**



A complex system is represented by **more than one model**, and the three above do not conflict. The **control input** and the **output of interest** decide which model is appropriate.

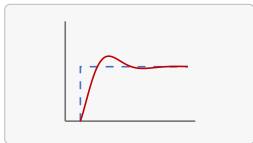
Four Stages of Analytical Study: Where This Course Lives



► Two Kinds of Analysis

- **Qualitative** first: is it stable? can the input steer every state? does the output reveal the state? $\rightarrow$ **yes or no**
- **Quantitative** next: how stable, how fast, how large is the response? $\rightarrow$ **a number or a curve**

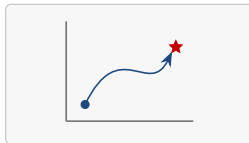
The qualitative answer comes first: no tuning rescues a plant that is not controllable.



what response?



stable?

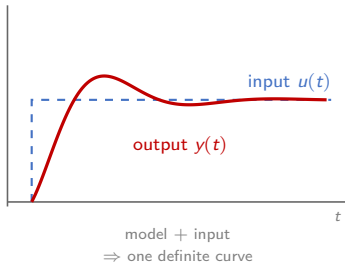


controllable?

Response and Stability: Two Analysis Questions

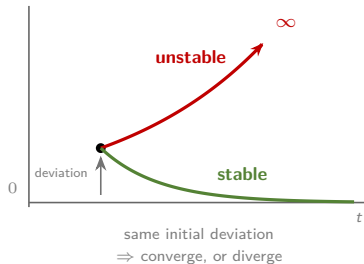
CONCEPT Response (quantitative)

- **Given:** a model *and* an input $u(t)$
- **Ask:** which output trajectory $y(t)$ follows? → answer is **a curve**



CONCEPT Stability (qualitative)

- **Given:** a model and an *initial deviation* from equilibrium
- **Ask:** does the state **converge** back, or **diverge**? → answer is **yes or no**



Quantitative analysis asks *how much*, and qualitative analysis asks *whether*. Design decisions rest on the second.

Interlude: When the Answer Is “Not Stable”

Tacoma Narrows Bridge, 1940

www.youtube.com/watch?v=mXTSnZgrfxM

Practical Engineering, *Why the Tacoma Narrows Bridge Collapsed*

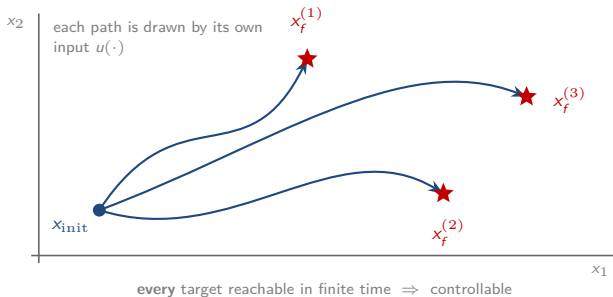
- ▶ The deck carried every load anyone had computed, so the **quantitative** answers were not wrong
- ▶ A small deviation grew on its own until the structure failed, so the **qualitative** answer was *not stable*
- ▶ The mechanism is **flutter**, not forced resonance: the airflow feeds energy back into the motion, so wind and deck form a **feedback loop** that turned unstable

Controllability: Can the Input Steer the State Anywhere?

CONCEPT Controllability (qualitative)

- **Given:** model, a start x_{init} , and *any* target x_f in the state space
- **Ask:** is there an input $u(\cdot)$ driving x_{init} to x_f in **finite time**, for *every* target? → answer is **yes or no**

Here $n = 2$: the state is the **vector** $x = (x_1, x_2)^T$, so each point of the plane is one state.



This course follows one qualitative property, **controllability**, asked repeatedly under **progressively weaker knowledge** of the model.

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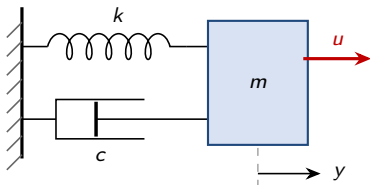
1.2 From One Plant to Networks

1.3 Book Outline

The Classical World: an *A* Whose Numbers We Know

- ▶ Classical control grew around a **single plant**: parameters measured on a test bench, identified from data, printed on a datasheet
- ▶ **Part I**: $\dot{x} = Ax + Bu$, $y = Cx$, with **every number in A trusted**
- ▶ Examples: mass–spring–damper, DC motor, with m, c, k, R, L all known

EXAMPLE Mass–Spring–Damper



y measured from equilibrium

$$m\ddot{y} + c\dot{y} + ky = u$$

$\ddot{y}$ acceleration · $\dot{y}$ velocity · y position

- ▶ Newton's second law: the spring pulls with ky and the damper with $c\dot{y}$
- ▶ **One second-order** equation in the single variable y

Measure m, c, k once on the bench and **every coefficient is fixed**. The next slide rewrites this one second-order equation as a **first-order** system, which is the form Part I works with.

From a Second-Order Equation to $\dot{x} = Ax + Bu$

► Step 1 Choose the state

$$x_1 = y \text{ (position)}, \quad x_2 = \dot{y} \text{ (velocity)}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

► Step 2 Write one first-order equation for each component

$$\dot{x}_1 = \dot{y} = x_2, \quad \dot{x}_2 = \ddot{y} = \frac{1}{m}(u - c\dot{y} - ky) = -\frac{k}{m}x_1 - \frac{c}{m}x_2 + \frac{1}{m}u$$

► Step 3 Stack the two equations

$$\underbrace{\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix}}_{\dot{x}} = \underbrace{\begin{bmatrix} 0 & 1 \\ -k/m & -c/m \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x + \underbrace{\begin{bmatrix} 0 \\ 1/m \end{bmatrix}}_B u, \quad y = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_C x$$

$$\dot{x} = Ax + Bu, \quad y = Cx$$

A the system · B the input · C the output, what the sensor reads

[The pattern] A linear plant of order n becomes n first-order equations in this way. Everything after this slide is stated for the triple (A, B, C) .

Networked Systems: Knowledge Inverted

- A network is **simple units joined by an interconnection structure** (a power grid, a robot team, a gene regulatory network), and a wiring diagram records that structure, so **which edges exist is known exactly**

$$\underbrace{\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix}}_{\dot{x}} = \underbrace{\begin{bmatrix} 0 & 0 & a_{13} & 0 \\ a_{21} & 0 & 0 & 0 \\ 0 & a_{32} & 0 & 0 \\ 0 & a_{42} & a_{43} & 0 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}}_x + \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_b u$$

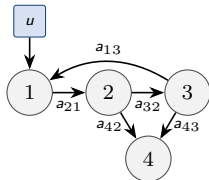
$$\dot{x}_1 = a_{13} x_3 + u$$

$$\dot{x}_2 = a_{21} x_1$$

$$\dot{x}_3 = a_{32} x_2$$

$$\dot{x}_4 = a_{42} x_2 + a_{43} x_3$$

x_i is
node i
 $a_{ij} \neq 0$ is an
edge $j \rightarrow i$

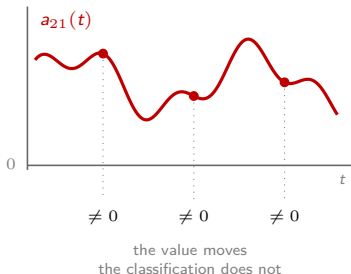


[The dictionary] $a_{ij} \neq 0$ exactly when the graph has an edge $j \rightarrow i$: matrix and picture are the **same object**. Can the five weights be trusted? That is the **next slide**.

Weights Drift: Numerical Verdicts Expire

CONCEPT Values Move, the Pattern Stays

- ▶ Even the “constants” are not constant: $a_{ij} = a_{ij}(t)$ with temperature, load, operating point
- ▶ So a rank verdict computed from *today's* numbers **expires tomorrow**, yet the edge is still there



▶ Keep Only the Classification

- ▶ Every entry is either **structurally zero** (no edge, known exactly) or a **free non-zero** parameter
- ▶ A matrix of that kind is a **pattern matrix**, and the network it describes is a **structured network**
- ▶ Erase the weights and one **pattern graph** serves every realization (notes Ch. 4)

[Premise of Part II] If the numbers in A cannot be trusted but the **zero/non-zero pattern** can, what remains of controllability?

Teaser: One Number Flips the Verdict

EXAMPLE Same graph, opposite verdicts (formally in week 3)

- ▶ Two systems of **identical pattern**: same nodes and same edges, with **one** edge weight differing

$$A_{\circ} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \text{ (uncontrollable)} \qquad A_{\bullet} = \begin{bmatrix} 1 & 2 \\ \textcolor{red}{3} & 2 \end{bmatrix} \text{ (controllable)}$$

- ▶ On a lab plant: a curiosity resolved by measurement
- ▶ On a network: **the weights are exactly what we do not know**. The **zeros are certain**, since a missing edge stays missing, and only the **non-zero** entries change

→ a numerical verdict holds for **today's numbers only**, not tomorrow's

What can the pattern alone certify? This question opens all of Part II. The answers are combinatorial: **read off a graph**.

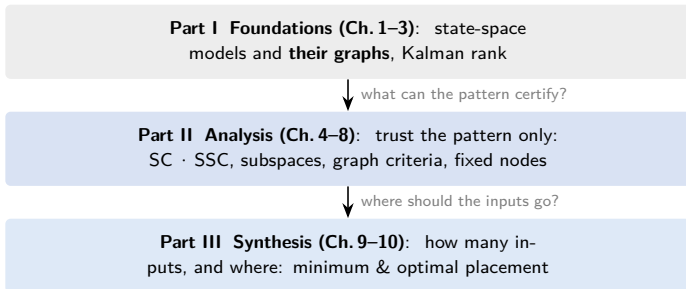
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Three Parts, One Chain of Questions



- ▶ Each chapter opens by answering **the question the previous one left**
- ▶ See the one-page roadmap figure in the Preface of the lecture notes

Two Symmetries Running Through the Book

► Symmetry 1: SC vs SSC

- **Structural (SC):** holds for *almost all* weights (generic)
- **Strong structural (SSC):** holds for *all* weights (worst case)

With one input $\mathcal{C}_F$ is square, so $\det \mathcal{C}_F$ is a polynomial in the free weights, and the two cases differ by how it can vanish.

chain $u \rightarrow 1 \rightarrow 2 \rightarrow 3$

$$A_F = \begin{bmatrix} 0 & 0 & 0 \\ a_{21} & 0 & 0 \\ 0 & a_{32} & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\det \mathcal{C}_F = a_{21}^2 a_{32}$$

a **product** of non-zero weights, never zero
 → **SSC**

flip two nodes, $b = (1, 1)^\top$

$$A_F = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\det \mathcal{C}_F = a_{21} + a_{22} - a_{11} - a_{12}$$

a **sum**, zero on a measure-zero set
 → **SC**, not SSC

► Symmetry 2: Algebra vs Graph

- Every condition stated in **both languages**: determinant · rank $\leftrightarrow$ stems · matchings · zero forcing

Prerequisites and Next Week

► Prerequisites (all reviewed in week 2)

- Undergraduate control (transfer functions, basic state space)
- Linear algebra (eigenvalues, rank) · basic programming (MATLAB)
- Graph theory is **not assumed** and is introduced within the course

► Next Week (Ch. 2: Foundations)

- State-space models and the **graph of a linear system** $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, the declared viewpoint of the book
- Linear-algebra toolkit · solutions and stability · Laplacian consensus dynamics

[Line of the day] “Every linear system is already a graph.
Part II merely **erases the numbers** on the edges.”